



Differentiating Emigration from Return Migration in Digital Trace Data

A case study of scholarly migration

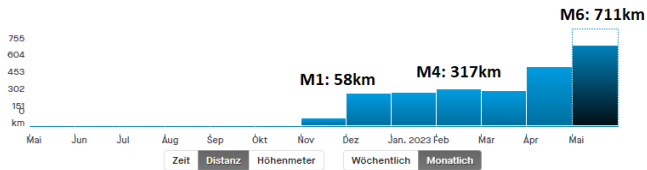
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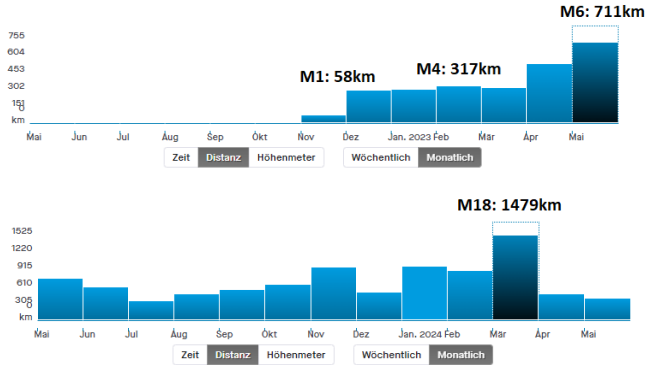
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- ▶ Most digital trace data does not include nationality.
- ▶ Left truncation issue in migration research, since we are uncertain about the country of origin.
- ▶ Identifying nationality enables differentiation between emigration and return migration.



Left truncation issue in digital trace data

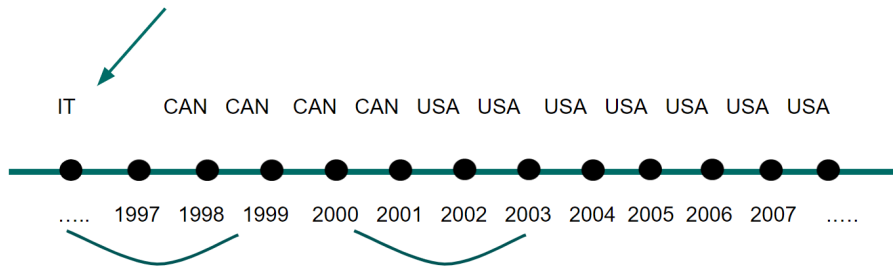


(figures from my coach's Strava profile)

- ▶ We detect the nationality with the least available data, full names, using Machine Learning with fasttext and char CNN.
- ▶ We use it instead of the country of academic origin in studying the migration of scholars.
- ▶ We exclude the main countries of immigration with a high diversity of scholars such as the US, the UK, Canada, and Australia from our test and evaluation steps.
- ▶ Once our model shows good performance, we use these data to analyze the composition of scholars' population.

How the ML method works

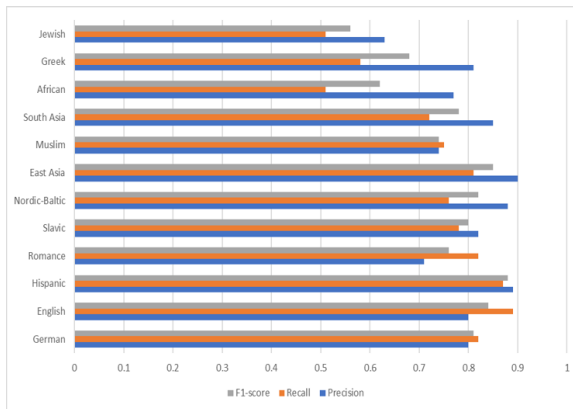
Maria's "full name", "**Maria Casnici**" allows a rough guess of her nationality



Maria's **migration history** can change!

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- This sunburst chart illustrates the global distribution of languages, categorized by language family, language, and country. The chart is divided into three concentric rings. The innermost ring represents the language family, the middle ring represents the specific language, and the outermost ring represents the country or region where the language is spoken. The size of each segment is proportional to the number of speakers.
- The chart shows that the Germanic language family (teal/green) is the largest, followed by the Romance language family (orange/brown). Other families include Slavic (dark brown), East Asian (light green), and South Asian (yellow). The chart also shows the distribution of languages within each family, such as German, French, Italian, Spanish, and English. The outer ring shows the countries where these languages are spoken, such as Germany, France, Italy, Spain, and the United Kingdom.

- Nested structure of regions and countries of origin based on the name in three levels (left) and performance of our Machine Learning model in level 1 in the most inner circle (right).



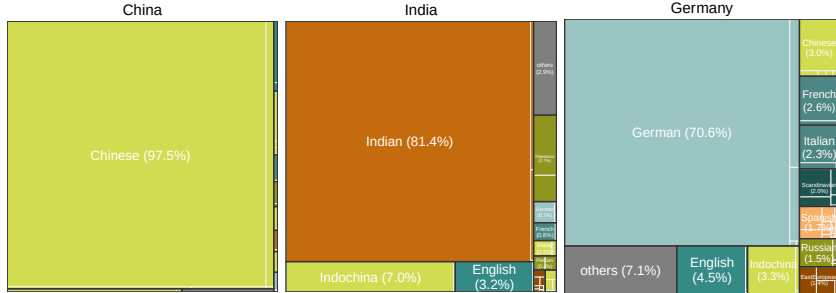
Name	Count
Wikipedia notable people's mobility	2,190,457
Nationality Estimate	696,136
WikiBio (Wikipedia Biography Dataset)	418,978
Wikipedia Person Dataset	355,468
Politicians on Wikipedia	54,877
Pantheon 1.0 dataset	16,071
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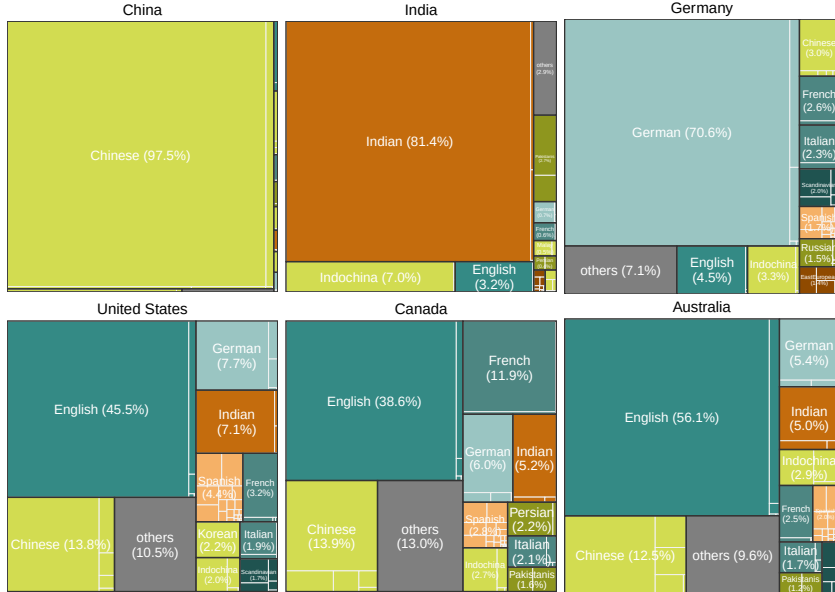
Analysis data is 25 million author names from Scopus

	Level 1		Level 2		Level 3	
	Balanced Accuracy (BA)	Weighted F1	BA	Weighted F1	BA	Weighted F1
IUSSP	72.7	73.4	66.5	66.9	57.1	57.1
Athletes	77.1	77.9	71.8	72.1	58.7	58.5

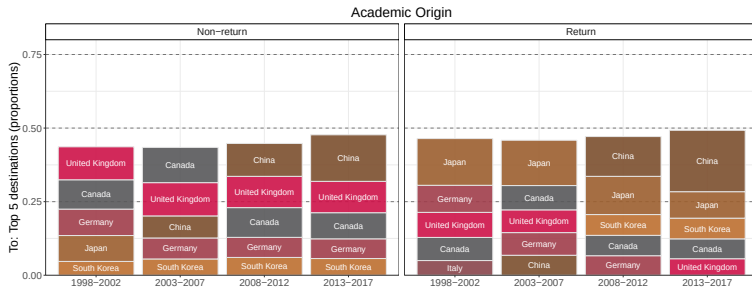
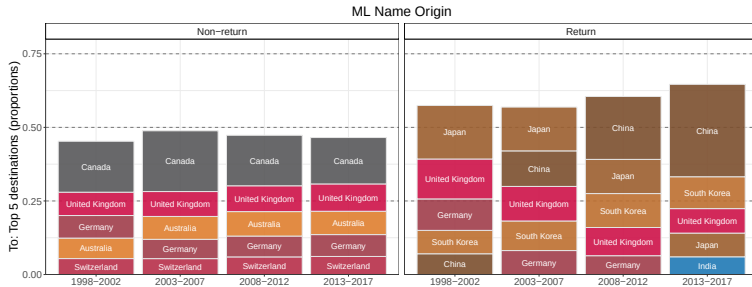
Population's Composition



Population's Composition

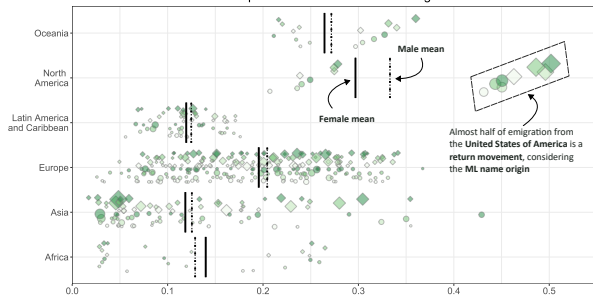


Top 5 destinations for emigrants from the United States

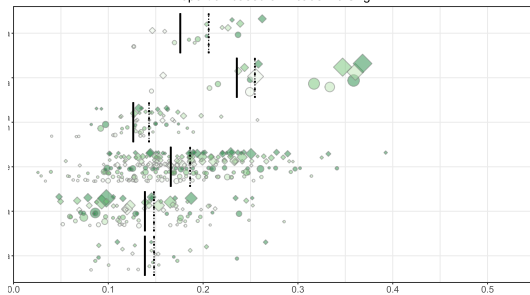


Emigration vs. Return Migration

Proportion based on ML Name Origin



Proportion based on Academic Origin



Period
 ○ 1998-2002 ● 2008-2012
 ● 2003-2007 ● 2013-2017

Scholar's Population (thousands)
 ○ 25 ○ 50 ○ 100 ○ 200

Gender
 ○ female
 ◇ male

- ▶ The literature reports high rates of emigration from the US to China (using the country of academic origin)
- ▶ Could part of this emigration be the return of Chinese graduate students and scholars to China? Could assigning nationality using ML help differentiate these cases?
- ▶ About 50% of emigration from the US was a return migration. 75% of emigration to China is a return migration. In 2013-2017, this increased to 87%. It was 26% and 43% using academic origin, respectively.
- ▶ Our ML model works best for names from some countries such as China, Japan, and South Korea.
- ▶ While we cannot differentiate second generation migrants, when a move is observed, we can say if it is a return migration.
- ▶ **Limitations**: Training data is over-represented by Western names. ML model's performance drops for Spanish and Southern American names that are hard to identify the country of origin.
- ▶ **Next**: show migration corridors where our model's results are the most reliable.

Thanks for your attention

Questions and comments are welcome!



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Examples from IUSSP's members directory



	Level 1		Level 2		Level 3	
Name	Reported	Predicted	Reported	Predicted	Reported	Predicted
Mohammad Jalal Abbasi-Shavazi	Muslim	Muslim	Persian	Persian	Iran	Iran
Emilio Zagheni	Romance	Romance	Italian	Italian	Italy	Italy
Guy Abel	English	Romance	English	French	United Kingdom	France
Mikko Myrskylä	Nordic-Baltic	Nordic-Baltic	Nordic	Nordic	Finland	Finland
Vladimir Canudas-Romo	Hispanic	Slavic	Spanish	Russian	Mexico	Russia
Midanna de Almada	African	Hispanic	South Africa	Spanish	Namibia	Portugal
Preeti Upadhyay	South Asia	South Asia	Indian	Indian	India	India
Louis Lohle-Tart	German	Romance	Dutch	French	Belgium	France
Nico Van Nimwegen	German	German	Dutch	Dutch	Netherlands	Netherlands
Andrew "Amos" Channon	English	English	English	English	United Kingdom	United Kingdom